Fitting and Interpreting Exponential Models

Precalculus

Fitting a model to two data points, interpreting its parameters, solving for time, and rewriting the base

Every model on this sheet has the form $y = a \cdot b^t$, where a is the value at $t = 0$ and b is the factor the quantity is multiplied by each time step. A growth rate r per step means $b = 1 + r$; a decay rate means $b = 1 - r$. To fit from two points, divide the two equations so a cancels; to solve for t, take logarithms of both sides; to change the base, use $b^t = e^{(\ln b)t}$.

Directions: Units and time origins are stated in each problem. Give exact values where they are exact, and round decimals to the precision the problem asks for. Show the equation you solved, not just the answer.

1. A streaming service models its subscribers by $S(t) = 320 \cdot (1.25)^t$ thousand subscribers, where t is the number of years since launch. State the number of subscribers at launch, and state the growth rate per year as a decimal.

Answer: ________________

2. A bacteria culture is modelled by $N(t) = a \cdot b^t$ cells, with t in hours. The culture holds 240 cells at $t = 0$ hours and 810 cells at $t = 3$ hours. Find the hourly growth factor b. (Divide the second equation by the first so that a cancels.)

Answer: ________________

3. The same culture is modelled by $N(t) = 240 \cdot (1.5)^{t}$ cells. At what time t, in hours, does the culture reach 1215 cells?

Answer: ________________

4. Rewrite the culture model $N(t) = 240 \cdot (1.5)^{t}$ in the continuous form $N(t) = 240 \cdot e^{kt}$. Give the exact value of k as a logarithm and then as a decimal rounded to three places.

Answer: ________________

5. A dose of medicine in a patient's bloodstream is modelled by $A(t) = 80 \cdot (0.85)^t$ milligrams, with t in hours.

(a) What decimal fraction of the medicine is lost each hour?

Answer: ________________

(b) How much medicine remains after 4 hours, in milligrams, to two decimal places?

Answer: ________________

6. Algae coverage on a pond is modelled by $C(t) = a \cdot b^t$ square metres, with t in weeks. Two measurements were recorded:

Week t	**Coverage** $C(t)$
1	36
4	121.5

(a) Find the weekly growth factor b.

Answer: ________________

(b) Find a, the coverage the model gives at $t = 0$.

Answer: ________________

7. Use the streaming model $S(t) = 320 \cdot (1.25)^{t}$ again. How long does the subscriber count take to double? Give the number of years to two decimal places. (Notice that the answer does not depend on 320.)

Answer: ________________

8. The streaming company wants to report growth per *month* instead of per year. Because 12 months make a year, the monthly factor b satisfies $b^{12} = 1.25$.

(a) Find the monthly growth factor to four decimal places.

Answer: ________________

(b) State the monthly growth rate as a decimal to four places.

Answer: ________________

9. Find and fix the mistake. A town's population is modelled by $P(t) = 500 \cdot (1.08)^t$ people, with t in years. A student reads the model and says: "the town gains 8 people every year".

(a) Compute the increase in population from $t = 0$ to $t = 1$.

Answer: ________________

(b) Compute the increase from $t = 5$ to $t = 6$, to two decimal places. Your two answers show why the student's reading is wrong.

Answer: ________________

10. A chemical sample decays exponentially: it measures 12 milligrams at $t = 0$ hours and 3 milligrams at $t = 6$ hours.

(a) Fit the model $M(t) = a \cdot b^t$: find the hourly decay factor b to four decimal places.

Answer: ________________

(b) Using that model, at what time does the sample fall to 1 milligram? Give the answer to two decimal places.

Answer: ________________ hours